

Precision Physics at the FCC-ee

Top FCNC Search at the FCC-ee

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Part I

Precision Physics at the FCC-ee

Chapter 1

Physics background and Precision Physics at the FCC-ee

1.1 FCC-ee as a precision machine

The Future Circular Collider programme is designed as an integrated precision-plus-energy-frontier strategy. The electron–positron stage, FCC-ee, would operate at several centre-of-mass energies chosen to maximise sensitivity to different sectors of the Standard Model: the Z pole, the WW threshold, the ZH Higgs stage, the $t\bar{t}$ threshold and a high-energy 365 GeV stage. The hadron stage, FCC-hh, then uses the same infrastructure to extend direct reach and high-energy precision. The relevant point for this talk is that rare top searches at FCC-ee are not disconnected from the rest of the programme: they exploit the same clean event reconstruction, vertexing, flavour tagging and global interpretation logic that define the Higgs and electroweak programmes [1, 3, 2].

Definition

A precision observable is an experimentally measurable quantity whose Standard-Model prediction and experimental uncertainty can be controlled well enough that small deviations become meaningful. Examples include m_Z , Γ_Z , $\sin^2 \theta_{\text{eff}}^\ell$, $\sigma(ZH)$, m_W , m_t and rare branching fractions such as $\text{Br}(t \rightarrow q\gamma)$.

In an effective-field-theory picture, the sensitivity of a precision observable to heavy new physics can be estimated schematically as

$$\frac{\delta\mathcal{O}}{\mathcal{O}_{\text{SM}}} \sim C_i \frac{v^2}{\Lambda^2}, \quad (1.1)$$

for observables controlled by electroweak symmetry breaking, or as

$$\frac{\delta\mathcal{O}}{\mathcal{O}_{\text{SM}}} \sim C_i \frac{E^2}{\Lambda^2}, \quad (1.2)$$

for high-energy amplitudes whose deviations grow with the scattering energy E . Here C_i is a dimensionless Wilson coefficient, $v \simeq 246$ GeV is the Higgs vacuum expectation value and Λ is the characteristic new-physics scale. A per-mille measurement therefore probes multi-TeV scales for order-one Wilson coefficients. This simple estimate is only a guide; real interpretations require correlations, theory uncertainties, detector effects and EFT validity checks.

Key message

The first scientific message of the talk is that precision measurements and rare processes are complementary discovery tools. Precision Higgs, electroweak and top measurements test correlated deviations, while top-FCNC searches test interactions that are essentially absent in the SM.

1.2 Why Z, WW, Higgs and top measurements are correlated

The FCC-ee programme is best understood as a network of measurements. The Z pole fixes electroweak vertices and pseudo-observables. The WW threshold tests the charged-current gauge sector and the W mass. The Higgs stage measures $\sigma(ZH)$ inclusively and exclusive Higgs decay rates. The top threshold and 365 GeV stage constrain m_t , Γ_t and top electroweak couplings. In the Standard Model Effective Field Theory (SMEFT), the same gauge-invariant operators can contribute to several of these sectors simultaneously:

$$\mathcal{L}_{\text{SMEFT}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{C_i^{(6)}}{\Lambda^2} \mathcal{O}_i^{(6)} + \sum_i \frac{C_i^{(8)}}{\Lambda^4} \mathcal{O}_i^{(8)} + \dots \quad (1.3)$$

A single-operator interpretation of one Higgs rate is usually too naive, because electroweak data may constrain the same direction. Conversely, Higgs rates can help interpret electroweak and top deviations. This is the reason for Slide 13: Higgs precision needs the Z and WW programmes.

1.3 Electroweak pseudo-observables and $Zf\bar{f}$ couplings

At the Z pole, the collider scans the resonance line shape and measures partial widths and asymmetries. The most important pseudo-observables include

$$m_Z, \quad \Gamma_Z, \quad \sigma_{\text{had}}^0, \quad R_\ell, \quad R_b, \quad R_c, \quad A_{\text{FB}}^{0,f}, \quad \sin^2 \theta_{\text{eff}}^\ell, \quad N_\nu. \quad (1.4)$$

The hadronic pole cross section is

$$\sigma_{\text{had}}^0 = \frac{12\pi}{m_Z^2} \frac{\Gamma_e \Gamma_{\text{had}}}{\Gamma_Z^2}. \quad (1.5)$$

The ratios

$$R_\ell = \frac{\Gamma_{\text{had}}}{\Gamma_{\ell\bar{\ell}}}, \quad R_b = \frac{\Gamma_{b\bar{b}}}{\Gamma_{\text{had}}}, \quad R_c = \frac{\Gamma_{c\bar{c}}}{\Gamma_{\text{had}}} \quad (1.6)$$

remove some common systematics and isolate flavour-dependent electroweak couplings. The effective number of light neutrinos can be expressed schematically as

$$N_\nu = \frac{\Gamma_{\text{inv}}}{\Gamma_{\text{SM}}^\nu}, \quad (1.7)$$

where Γ_{inv} is the invisible Z width after subtracting the visible partial widths. Deviations in N_ν may arise from new invisible particles or modified neutrino couplings, but precision interpretation requires careful control of the full Z-width fit.

1.3.1 Chiral couplings

The Z boson couples differently to left- and right-handed fermions. Define the chirality projectors

$$P_L = \frac{1 - \gamma_5}{2}, \quad P_R = \frac{1 + \gamma_5}{2}, \quad f_L = P_L f, \quad f_R = P_R f. \quad (1.8)$$

A useful convention is

$$\mathcal{L}_{Zff} = \frac{g}{c_W} Z_\mu \bar{f} \gamma^\mu (g_L^f P_L + g_R^f P_R) f, \quad (1.9)$$

with

$$g_L^f = T_3^f - Q_f \sin^2 \theta_{\text{eff}}^f, \quad g_R^f = -Q_f \sin^2 \theta_{\text{eff}}^f. \quad (1.10)$$

Right-handed SM fermions are singlets under $SU(2)_L$, so $T_3 = 0$ for their weak isospin. Left-handed fermions sit in $SU(2)_L$ doublets and have $T_3 = \pm 1/2$. The vector and axial combinations are conventionally related by

$$g_V^f = g_L^f + g_R^f, \quad g_A^f = g_L^f - g_R^f, \quad (1.11)$$

up to global normalisation conventions. Partial widths and asymmetries probe different combinations:

$$\Gamma_f \propto (g_L^f)^2 + (g_R^f)^2, \quad A_f = \frac{(g_L^f)^2 - (g_R^f)^2}{(g_L^f)^2 + (g_R^f)^2}, \quad (1.12)$$

$$A_{\text{FB}}^{0,f} = \frac{3}{4} A_e A_f. \quad (1.13)$$

This is why rates, line-shape information and asymmetries together separate left- and right-handed shifts.

1.4 WW threshold, gauge structure and aTGCs

Near the WW threshold the cross section $e^+e^- \rightarrow W^+W^-$ is very sensitive to m_W and Γ_W . Above threshold, differential angular distributions are sensitive to the non-Abelian electroweak gauge structure. A common anomalous triple-gauge-coupling parameterisation is

$$\mathcal{L}_{WWV} = -ig_{WWV} \left[g_1^V \left(W_{\mu\nu}^+ W^{-\mu} V^\nu - W_{\mu\nu}^- W^{+\mu} V^\nu \right) + \kappa_V W_\mu^+ W_\nu^- V^{\mu\nu} \right] \quad (1.14)$$

$$+ \frac{\lambda_V}{m_W^2} W_{\mu\nu}^+ W^{-\nu\rho} V_\rho{}^\mu, \quad V = \gamma, Z. \quad (1.15)$$

At tree level in the SM,

$$g_1^\gamma = g_1^Z = \kappa_\gamma = \kappa_Z = 1, \quad \lambda_\gamma = \lambda_Z = 0. \quad (1.16)$$

The slide uses $\delta g_{1,Z}$, $\delta\kappa_\gamma$ and λ_Z as the key deviations. Here $\delta g_{1,Z}$ modifies the overall WWZ structure, $\delta\kappa_\gamma$ acts like an anomalous magnetic-dipole contribution to $WW\gamma$, and λ_Z is a momentum-dependent higher-derivative interaction. In SMEFT, these parameters are correlated with Higgs-gauge interactions such as HZZ and HWW .

1.5 Higgs recoil method and absolute coupling extraction

At $\sqrt{s} \simeq 240$ GeV, the process

$$e^+e^- \rightarrow ZH \quad (1.17)$$

operates near the Higgsstrahlung cross-section maximum. If the Z boson is reconstructed, for example through $Z \rightarrow \mu^+\mu^-$ or $Z \rightarrow e^+e^-$, the recoil mass is

$$m_{\text{recoil}}^2 = (p_{e^+} + p_{e^-} - p_Z)^2 \quad (1.18)$$

$$= s + m_Z^2 - 2\sqrt{s}E_Z, \quad (1.19)$$

where E_Z is the reconstructed Z-boson energy. The key point is that this observable does not require reconstructing the Higgs decay. Therefore invisible, exotic or untagged Higgs decays still contribute to the inclusive ZH peak. The inclusive cross section scales as

$$\sigma(ZH) \propto g_{HZZ}^2 \equiv \kappa_Z^2 (g_{HZZ}^{\text{SM}})^2. \quad (1.20)$$

This gives the absolute Higgs normalisation that is missing at hadron colliders.

Exclusive Higgs decays then measure products of production and decay factors:

$$\sigma(ZH) \text{Br}(H \rightarrow X) \propto \frac{g_{HZZ}^2 g_{HXX}^2}{\Gamma_H}, \quad (1.21)$$

while WW-fusion gives

$$\sigma(\nu\bar{\nu}H) \text{Br}(H \rightarrow X) \propto \frac{g_{HWW}^2 g_{HXX}^2}{\Gamma_H}. \quad (1.22)$$

Combining recoil, exclusive decays and vector-boson fusion gives direct access to absolute couplings and the total width Γ_H .

1.5.1 Exclusive Higgs decay channels

Mode	Coupling information	Physics role
$H \rightarrow b\bar{b}$	bottom Yukawa κ_b	Dominant Higgs decay; central for absolute coupling fits and flavour-tagging performance.
$H \rightarrow c\bar{c}$	charm Yukawa κ_c	Direct second-generation quark coupling; requires charm tagging and control of $b/c/g$ separation.
$H \rightarrow gg$	effective Hgg modifier κ_g	Loop-induced in the SM; sensitive to top loops, QCD and possible coloured BSM states.
$H \rightarrow \tau^+\tau^-$	tau Yukawa κ_τ	Tests third-generation lepton mass generation and tau reconstruction.
$H \rightarrow WW^*$	HWW coupling κ_W	Important width handle; one W is off shell for a 125 GeV Higgs.
$H \rightarrow ZZ^*$	HZZ coupling κ_Z	Clean gauge-coupling channel and width handle.
$H \rightarrow \gamma\gamma$	effective $H\gamma\gamma$ modifier κ_γ	Rare loop-induced mode; FCC-hh statistics strongly improve precision.
$H \rightarrow Z\gamma$	effective $HZ\gamma$ modifier $\kappa_{Z\gamma}$	Rare loop-induced mode with strong BSM sensitivity.
$H \rightarrow \mu^+\mu^-$	muon Yukawa κ_μ	Second-generation lepton Yukawa; benefits strongly from FCC-hh.

1.6 Higgs coupling modifiers and the κ -3 framework

In the coupling-modifier framework,

$$\kappa_i = \frac{g_i}{g_i^{\text{SM}}}, \quad (1.23)$$

for tree-level couplings, while loop-induced processes are assigned effective modifiers such as κ_g , κ_γ and $\kappa_{Z\gamma}$. A production-times-decay rate can be written as

$$(\sigma \cdot \text{Br})(i \rightarrow H \rightarrow f) = \sigma_i \frac{\Gamma_f}{\Gamma_H}. \quad (1.24)$$

In the κ framework,

$$(\sigma \cdot \text{Br})(i \rightarrow H \rightarrow f) = \left(\sigma_i^{\text{SM}} \kappa_i^2 \right) \frac{\Gamma_f^{\text{SM}} \kappa_f^2}{\Gamma_H^{\text{SM}} \kappa_H^2}, \quad (1.25)$$

so that the signal strength is

$$\mu_i^f = \frac{(\sigma \cdot \text{Br})}{(\sigma \cdot \text{Br})_{\text{SM}}} = \frac{\kappa_i^2 \kappa_f^2}{\kappa_H^2}. \quad (1.26)$$

The total-width modifier is approximated by

$$\kappa_H^2 = \sum_j \kappa_j^2 \text{Br}_j^{\text{SM}}, \quad (1.27)$$

with the sum over all relevant SM decay modes. If invisible or untagged modes are allowed, the κ -3 framework often uses

$$\bar{\kappa}_H^2 = \frac{\kappa_H^2}{1 - B_{\text{inv}} - B_{\text{und}}}, \quad \Gamma_H = \bar{\kappa}_H^2 \Gamma_H^{\text{SM}}. \quad (1.28)$$

The condition $\kappa_V \leq 1$ is sometimes imposed in hadron-collider fits to stabilise the width interpretation. At FCC-ee, inclusive ZH recoil strongly reduces this degeneracy by directly measuring $\sigma(ZH)$.

Caution

The κ framework is useful for explaining Higgs-rate measurements, but it is not a gauge-invariant EFT. SMEFT is required when one wants to connect Higgs, electroweak, diboson and top observables consistently.

1.7 SMEFT interpretation of Higgs, EW and top data

The SMEFT operators most relevant for the motivation slides include bosonic operators and current operators. Examples are

$$\mathcal{O}_{HWB} = H^\dagger \sigma^a H W_{\mu\nu}^a B^{\mu\nu}, \quad (1.29)$$

$$\mathcal{O}_{HD} = \left| H^\dagger D_\mu H \right|^2, \quad (1.30)$$

$$\mathcal{O}_{H\ell}^{(1)} = (H^\dagger i \overleftrightarrow{D}_\mu H) (\bar{\ell} \gamma^\mu \ell), \quad (1.31)$$

$$\mathcal{O}_{H\ell}^{(3)} = (H^\dagger i \overleftrightarrow{D}_\mu^a H) (\bar{\ell} \sigma^a \gamma^\mu \ell), \quad (1.32)$$

$$\mathcal{O}_{He} = (H^\dagger i \overleftrightarrow{D}_\mu H) (\bar{e} \gamma^\mu e). \quad (1.33)$$

These can shift Zee couplings, affect HZZ and $HZee$ contact interactions and correlate with diboson and Higgs production. Bosonic operators such as $\mathcal{O}_{HW}$, $\mathcal{O}_{HB}$ and $\mathcal{O}_W$ affect gauge-boson kinetic/momentum structures and anomalous triple-gauge couplings. This is the origin of the chords in the circo plot: the plot visualises correlations among fitted effective couplings, not independent measurements.

1.8 Top-quark precision at FCC-ee

The top threshold programme measures the final state

$$e^+ e^- \rightarrow W^+ b W^- \bar{b}, \quad (1.34)$$

with resonant and non-resonant contributions. The threshold line shape can be written schematically as

$$\sigma_{WbWb} = \sigma_{WbWb}(\sqrt{s}; m_t, \Gamma_t, \alpha_s, y_t). \quad (1.35)$$

The horizontal position of the line shape is sensitive to the top mass, the width controls the sharpness, α_s affects the QCD potential, and the top Yukawa coupling produces smaller but relevant effects through Higgs exchange. A threshold mass such as the potential-subtracted mass m_t^{PS} is used to avoid the pole-mass ambiguity.

Above threshold, the process

$$e^+ e^- \rightarrow \gamma^*, Z^* \rightarrow t \bar{t} \quad (1.36)$$

probes the top electroweak vertex. A compact notation is

$$\mathcal{L}_{t\bar{t}Z} = \frac{g}{2c_W} \bar{t} \gamma^\mu (g_V^t - g_A^t \gamma_5) t Z_\mu. \quad (1.37)$$

Angular distributions and forward–backward asymmetries are essential to separate chiral structures.

1.9 FCNC in the Standard Model and beyond

Flavour-changing neutral currents (FCNCs) such as $t \rightarrow q\gamma$, $t \rightarrow qZ$, $t \rightarrow qH$ and $t \rightarrow qg$ are absent at tree level in the SM. They appear only through loops and are strongly suppressed by the Glashow–Iliopoulos–Maiani mechanism. The expected branching ratios are tiny, often around 10^{-12}

or below depending on the channel. Therefore any observable rate far above the SM expectation would strongly indicate BSM dynamics [7].

The talk focuses on $tq\gamma$ and tqZ because these interactions induce clean single-top production in e^+e^- collisions:

$$e^+e^- \rightarrow \gamma^*/Z^* \rightarrow t\bar{q}, \bar{t}q, \quad q = u, c. \quad (1.38)$$

The produced top decays through the SM mode

$$t \rightarrow Wb, \quad (1.39)$$

which leads to semileptonic and fully hadronic signatures.

1.10 Effective Lagrangian for top-FCNC interactions

The anomalous-coupling framework used in the slides follows the common Aguilar-Saavedra normalisation [8]. The interaction is

$$\mathcal{L}_{\text{FCNC}}^{tqX} = \sum_{q=u,c} \left[\frac{g_s}{2m_t} \bar{q} \lambda^a \sigma^{\mu\nu} (\zeta_{qt}^L P_L + \zeta_{qt}^R P_R) t G_{\mu\nu}^a - \frac{1}{\sqrt{2}} \bar{q} (\eta_{qt}^L P_L + \eta_{qt}^R P_R) t H \right. \quad (1.40)$$

$$\left. - \frac{g_W}{2c_W} \bar{q} \gamma^\mu (X_{qt}^L P_L + X_{qt}^R P_R) t Z_\mu + \frac{g_W}{4c_W m_Z} \bar{q} \sigma^{\mu\nu} (\kappa_{qt}^L P_L + \kappa_{qt}^R P_R) t Z_{\mu\nu} \right. \quad (1.41)$$

$$\left. + \frac{e}{2m_t} \bar{q} \sigma^{\mu\nu} (\lambda_{qt}^L P_L + \lambda_{qt}^R P_R) t A_{\mu\nu} \right] + \text{h.c.} \quad (1.42)$$

Here $q = u, c$, g_s is the strong coupling, g_W the weak $SU(2)_L$ coupling, e the electromagnetic coupling, $c_W = \cos \theta_W$, λ^a are colour generators and $\sigma^{\mu\nu} = i[\gamma^\mu, \gamma^\nu]/2$. The field strengths are $G_{\mu\nu}^a$, $Z_{\mu\nu}$ and $A_{\mu\nu}$. The coefficients ζ , η , X , κ and λ parameterise, respectively, tqg , tqH , vector tqZ , dipole tqZ and dipole $tq\gamma$ interactions.

Caution

The benchmark assumption $\lambda^L = \lambda^R$ or $\kappa^L = \kappa^R$ is a practical signal-generation choice. It is not a model-independent statement. Different BSM models can generate strongly chiral patterns.

1.11 Translation between anomalous couplings and branching ratios

The branching ratio is defined as

$$\text{Br}(t \rightarrow qX) = \frac{\Gamma(t \rightarrow qX)}{\Gamma_t^{\text{SM}} + \sum_X \Gamma(t \rightarrow qX)} \simeq \frac{\Gamma(t \rightarrow qX)}{\Gamma_t^{\text{SM}}}, \quad (1.43)$$

where the approximation holds for small FCNC widths. In the convention of Eq. (1.42), neglecting the light-quark mass, the photon-dipole partial width is commonly written as

$$\Gamma(t \rightarrow q\gamma) = \frac{\alpha}{4} m_t \left(|\lambda_{qt}^L|^2 + |\lambda_{qt}^R|^2 \right). \quad (1.44)$$

For the Z dipole, with $r_Z = m_Z^2/m_t^2$,

$$\Gamma(t \rightarrow qZ)_\kappa = \frac{\alpha}{32s_W^2 c_W^2} \frac{m_t^3}{m_Z^2} (1 - r_Z)^2 (2 + r_Z) \left(|\kappa_{qt}^L|^2 + |\kappa_{qt}^R|^2 \right). \quad (1.45)$$

For the symmetric benchmark $\lambda^L = \lambda^R = \lambda$ and $\kappa^L = \kappa^R = \kappa$,

$$\text{Br}(t \rightarrow q\gamma) \simeq \frac{\alpha m_t}{2\Gamma_t^{\text{SM}}} |\lambda_{qt}|^2, \quad (1.46)$$

$$\text{Br}(t \rightarrow qZ)_\kappa \simeq \frac{\alpha}{16s_W^2 c_W^2} \frac{m_t^3}{m_Z^2 \Gamma_t^{\text{SM}}} (1 - r_Z)^2 (2 + r_Z) |\kappa_{qt}|^2. \quad (1.47)$$

The numerical translation must use the same coupling normalisation as the UFO model used for event generation.

1.12 Signal processes at 240 and 365 GeV

The signal process is

$$e^+e^- \rightarrow t\bar{q} + \bar{t}q, \quad (1.48)$$

mediated by the anomalous $tq\gamma$ and tqZ couplings. At 240 GeV, the stage is optimised for Higgs studies but also provides clean single-top kinematics. At 365 GeV, there is more phase space and harder objects, but also genuine top backgrounds. The benchmark luminosities used in the slides are 5 ab^{-1} at 240 GeV and 1.5 ab^{-1} at 365 GeV.

1.13 Semileptonic and fully hadronic final states

The semileptonic topology is

$$e^+e^- \rightarrow tq \rightarrow Wbq \rightarrow \ell\nu bq, \quad (1.49)$$

with one isolated electron or muon, missing momentum from the neutrino, one b -tagged jet and one energetic light jet. It is clean but has a smaller branching fraction. The fully hadronic topology is

$$e^+e^- \rightarrow tq \rightarrow Wbq \rightarrow q\bar{q}'bq, \quad (1.50)$$

with four exclusive jets. It has a larger branching fraction but suffers from larger diboson backgrounds and combinatorics. A typical assignment variable is

$$\chi^2 = \frac{(m_{jj} - m_W)^2}{\sigma_W^2} + \frac{(m_{bjj} - m_t)^2}{\sigma_t^2}, \quad (1.51)$$

where the jet combination minimising χ^2 is chosen as the top candidate.

1.14 Backgrounds, jet clustering and flavour tagging

The dominant backgrounds are WW , ZZ , ZH , $e\gamma$ initiated processes, and at 365 GeV also $t\bar{t}$ and single-top-like final states. The analysis adopts the ee - k_T Durham algorithm in exclusive two-jet or four-jet modes. E-scheme recombination means four-momenta are added directly. The slides quote a flavour-tagging working point $\epsilon_b = 80\%$ and $\epsilon_c = 10\%$. In the FCNC selection, b tagging identifies the top decay, while light-jet rejection and charm/light separation control backgrounds.

Discriminating variables include reconstructed masses M_W and M_{top} , the b -jet momentum p_b , the light-jet momentum p_{light} , missing momentum, angular separations and event activity. If H_T is used, define it in the analysis note as the scalar sum of visible transverse momenta or the corresponding FCC-ee event-activity proxy:

$$H_T = \sum_i p_{T,i}. \quad (1.52)$$

The usual hadron-collider angular distance is

$$\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}, \quad (1.53)$$

although in e^+e^- analyses ordinary opening angles are often equally transparent.

1.15 Statistical analysis and CL_s extraction

A generic binned likelihood for the combined analysis is

$$\mathcal{L}(\mu, \boldsymbol{\theta}) = \prod_{c,b} \text{Pois}(n_{cb} | \mu s_{cb}(\boldsymbol{\theta}) + b_{cb}(\boldsymbol{\theta})) \prod_k \pi_k(\theta_k), \quad (1.54)$$

where c labels channels, b labels histogram bins, n_{cb} are observed or Asimov counts, s_{cb} and b_{cb} are signal and background templates, μ is the signal-strength parameter, and θ_k are nuisance parameters. The CL_s ratio is conventionally

$$CL_s = \frac{p_{s+b}}{1 - p_b}, \quad (1.55)$$

with the exclusion set at $CL_s = 0.05$ for a 95% CL upper limit. In projected studies the observed dataset is often an Asimov dataset, so the quoted limit is an expected sensitivity.

1.16 Interpretation of projected limits

The final tables quote limits on $\text{Br}(t \rightarrow c\gamma)$ and $\text{Br}(t \rightarrow cZ)$ rather than directly on λ and κ . This is useful for comparing with LHC results and with other future-collider projections. The lepton channel gives the best sensitivity in the present benchmark because the final state is cleaner and the backgrounds are more controllable. The hadronic channel still contributes because its branching fraction is larger and it probes a complementary topology. The results should be described as study-level projections, subject to the stated luminosities, detector assumptions, signal model and systematic-treatment assumptions.

Part II

Technical appendices

Appendix A

Coupling and notation dictionary

Symbol	Meaning
$\sqrt{s}$	Centre-of-mass energy of the e^+e^- collision.
m_Z, Γ_Z	Z-boson mass and total width.
m_W, Γ_W	W-boson mass and total width.
$\sin^2 \theta_{\text{eff}}^\ell$	Effective weak mixing angle measured in leptonic asymmetries.
R_b	Ratio $\Gamma_{b\bar{b}}/\Gamma_{\text{had}}$.
N_ν	Effective number of light neutrino species inferred from invisible Z width.
$A_{\text{FB}}^{0,f}$	Pole forward–backward asymmetry for fermion f .
P_L, P_R	Chiral projectors $(1 \mp \gamma_5)/2$.
g_L^f, g_R^f	Left- and right-handed Z couplings to fermion f .
g_V^f, g_A^f	Vector and axial-vector Z couplings.
$\delta g_{1,Z}$	Deviation in the WWZ triple-gauge coupling.
$\delta \kappa_\gamma$	Deviation in the $WW\gamma$ magnetic-dipole-like coupling.
λ_Z	Higher-derivative anomalous WWZ coupling.
κ_Z, κ_W	Higgs coupling modifiers for HZZ and HWW .
$\kappa_b, \kappa_c, \kappa_t$	Higgs Yukawa modifiers for bottom, charm and top.
$\kappa_g, \kappa_\gamma, \kappa_{Z\gamma}$	Effective loop-induced Higgs modifiers.
B_{inv}	Invisible Higgs branching ratio.
B_{und} or B_{unt}	Undetected/untagged Higgs branching ratio.
Γ_H	Total Higgs width.
m_t^{PS}	Potential-subtracted top mass used for threshold fits.
ζ_{qt}	Chromomagnetic tqg FCNC coefficient.
η_{qt}	Scalar tqH FCNC coefficient.
X_{qt}	Vector tqZ FCNC coefficient.
κ_{qt}	Dipole tqZ FCNC coefficient.
λ_{qt}	Dipole $tq\gamma$ FCNC coefficient.
ϵ_b, ϵ_c	b -tagging efficiency and charm-tagging/mistag working-point number as used in the analysis.
CL_s	Modified frequentist confidence-level ratio used for expected upper limits.

Appendix B

Electroweak formula sheet

$$\sigma_{\text{had}}^0 = \frac{12\pi}{m_Z^2} \frac{\Gamma_e \Gamma_{\text{had}}}{\Gamma_Z^2}, \quad R_\ell = \frac{\Gamma_{\text{had}}}{\Gamma_{\ell\ell}}, \quad R_b = \frac{\Gamma_{b\bar{b}}}{\Gamma_{\text{had}}}, \quad (\text{B.1})$$

$$A_f = \frac{(g_L^f)^2 - (g_R^f)^2}{(g_L^f)^2 + (g_R^f)^2}, \quad A_{\text{FB}}^{0,f} = \frac{3}{4} A_e A_f, \quad N_\nu = \frac{\Gamma_{\text{inv}}}{\Gamma_\nu^{\text{SM}}}. \quad (\text{B.2})$$

These formulae are sufficient for the oral discussion. A full FCC electroweak fit also includes QED radiator functions, beam-energy calibration, luminosity theory and correlations among pseudo-observables.

Appendix C

Higgs recoil and coupling-extraction formula sheet

$$m_{\text{recoil}}^2 = s + m_Z^2 - 2\sqrt{s}E_Z, \quad (\text{C.1})$$

$$\sigma(ZH) \propto \kappa_Z^2, \quad (\text{C.2})$$

$$\sigma(ZH)\text{Br}(H \rightarrow X) \propto \frac{\kappa_Z^2 \kappa_X^2}{\Gamma_H}, \quad (\text{C.3})$$

$$\sigma(\nu\bar{\nu}H)\text{Br}(H \rightarrow X) \propto \frac{\kappa_W^2 \kappa_X^2}{\Gamma_H}, \quad (\text{C.4})$$

$$\mu_i^f = \frac{\kappa_i^2 \kappa_f^2}{\kappa_H^2}, \quad (\text{C.5})$$

$$\bar{\kappa}_H^2 = \frac{\kappa_H^2}{1 - B_{\text{inv}} - B_{\text{und}}}. \quad (\text{C.6})$$

Appendix D

SMEFT operator map

Operator	Main effect
$\mathcal{O}_{HWB}$	Shifts electroweak mixing and correlates EWPOs with Higgs-gauge couplings.
$\mathcal{O}_{HD}$	Affects custodial-sensitive electroweak observables and Higgs-gauge interactions.
$\mathcal{O}_{H\ell}^{(1)}, \mathcal{O}_{H\ell}^{(3)}$	Shift lepton electroweak currents, including Zee and $We\nu$.
$\mathcal{O}_{He}$	Shifts right-handed electron coupling to the Z.
$\mathcal{O}_{HW}, \mathcal{O}_{HB}$	Modify Higgs-gauge and diboson interactions.
$\mathcal{O}_W$	Produces momentum-dependent triple-gauge-coupling deviations.
Top dipoles and currents	Affect top EW couplings and loop-induced Higgs/top correlations.

Appendix E

Top-FCNC branching-ratio formula sheet

For the convention used in the slides,

$$\Gamma(t \rightarrow q\gamma) = \frac{\alpha}{4} m_t (|\lambda_{qt}^L|^2 + |\lambda_{qt}^R|^2), \quad (\text{E.1})$$

$$\Gamma(t \rightarrow qZ)_\kappa = \frac{\alpha}{32s_W^2 c_W^2} \frac{m_t^3}{m_Z^2} (1 - r_Z)^2 (2 + r_Z) (|\kappa_{qt}^L|^2 + |\kappa_{qt}^R|^2), \quad (\text{E.2})$$

$$\text{Br}(t \rightarrow qX) \simeq \Gamma(t \rightarrow qX)/\Gamma_t^{\text{SM}}, \quad r_Z = m_Z^2/m_t^2. \quad (\text{E.3})$$

Caution

These expressions must not be combined with a differently normalised FCNC Lagrangian. If another UFO model is used, the conversion between coupling limits and branching ratios must be rederived or checked against the model documentation.

Appendix F

Analysis checklist

1. Generate FCNC signal with one anomalous coupling switched on at a time.
2. Simulate detector response in the official FCC chain or a documented Delphes/IDEA approximation.
3. Reconstruct exclusive two-jet and four-jet topologies with the selected ee-Durham algorithm.
4. Apply lepton, jet, angular and flavour-tagging selections.
5. Build signal and background templates for discriminating variables.
6. Construct channel-wise likelihoods and combine channels/energies.
7. Translate signal-strength limits into branching-ratio limits using the same Lagrangian normalisation.
8. Repeat with systematic uncertainties and possibly multivariate discriminants for a final public analysis.

Appendix G

Extended formula compendium

This appendix collects compact formulae that are useful during Q&A. They are intentionally redundant with the main text, because they provide a quick technical lookup while presenting.

G.1 Kinematics and event reconstruction

$$s = (p_{e^+} + p_{e^-})^2, \quad E_{\text{beam}} = \sqrt{s}/2, \quad (\text{G.1})$$

$$p_{\text{miss}}^\mu = p_{e^+}^\mu + p_{e^-}^\mu - \sum_i p_i^\mu, \quad E_{\text{miss}} = p_{\text{miss}}^0, \quad (\text{G.2})$$

$$m_{ij}^2 = (p_i + p_j)^2, \quad m_{bjj}^2 = (p_b + p_{j_1} + p_{j_2})^2, \quad (\text{G.3})$$

$$H_T = \sum_i p_{T,i}, \quad \Delta R_{ij} = \sqrt{(\eta_i - \eta_j)^2 + (\phi_i - \phi_j)^2}, \quad (\text{G.4})$$

$$\cos \theta_{ij} = \frac{\vec{p}_i \cdot \vec{p}_j}{|\vec{p}_i| |\vec{p}_j|}, \quad \chi_W^2 = \frac{(m_{jj} - m_W)^2}{\sigma_W^2}, \quad (\text{G.5})$$

$$\chi_t^2 = \frac{(m_{bjj} - m_t)^2}{\sigma_t^2}, \quad \chi_{\text{tot}}^2 = \chi_W^2 + \chi_t^2. \quad (\text{G.6})$$

G.2 Electroweak couplings

$$Q_e = -1, \quad T_3^{eL} = -\frac{1}{2}, \quad T_3^{\nu L} = +\frac{1}{2}, \quad (\text{G.7})$$

$$g_L^f = T_3^f - Q_f s_{\text{eff}}^2, \quad g_R^f = -Q_f s_{\text{eff}}^2, \quad (\text{G.8})$$

$$g_V^f = g_L^f + g_R^f, \quad g_A^f = g_L^f - g_R^f, \quad (\text{G.9})$$

$$\Gamma_f = N_c^f \frac{G_F m_Z^3}{6\sqrt{2}\pi} \left[(g_V^f)^2 + (g_A^f)^2 \right] (1 + \delta_f), \quad (\text{G.10})$$

$$A_f = \frac{2g_V^f g_A^f}{(g_V^f)^2 + (g_A^f)^2}, \quad A_{\text{FB}}^{0,f} = \frac{3}{4} A_e A_f, \quad (\text{G.11})$$

$$R_b = \Gamma_b / \Gamma_{\text{had}}, \quad R_c = \Gamma_c / \Gamma_{\text{had}}, \quad (\text{G.12})$$

$$\Gamma_Z = \sum_f \Gamma_f + \Gamma_{\text{inv}}, \quad \Gamma_{\text{inv}} = N_\nu \Gamma_\nu^{\text{SM}}. \quad (\text{G.13})$$

G.3 Higgs rates

$$\sigma_{ZH} = \kappa_Z^2 \sigma_{ZH}^{\text{SM}}, \quad \Gamma_X = \kappa_X^2 \Gamma_X^{\text{SM}}, \quad (\text{G.14})$$

$$\text{Br}_X = \Gamma_X / \Gamma_H, \quad \mu_i^f = \frac{\sigma_i \text{Br}_f}{(\sigma_i \text{Br}_f)_{\text{SM}}}, \quad (\text{G.15})$$

$$\mu_i^f = \frac{\kappa_i^2 \kappa_f^2}{\kappa_H^2}, \quad \kappa_H^2 = \sum_j \kappa_j^2 \text{Br}_j^{\text{SM}}, \quad (\text{G.16})$$

$$\bar{\kappa}_H^2 = \frac{\kappa_H^2}{1 - B_{\text{inv}} - B_{\text{und}}}, \quad \Gamma_H = \bar{\kappa}_H^2 \Gamma_H^{\text{SM}}, \quad (\text{G.17})$$

$$\sigma(ZH) \text{Br}_{b\bar{b}} \propto \frac{\kappa_Z^2 \kappa_b^2}{\Gamma_H}, \quad \sigma(ZH) \text{Br}_{ZZ^*} \propto \frac{\kappa_Z^4}{\Gamma_H}, \quad (\text{G.18})$$

$$\sigma(\nu\bar{\nu}H) \text{Br}_X \propto \frac{\kappa_W^2 \kappa_X^2}{\Gamma_H}, \quad m_{\text{recoil}}^2 = s + m_Z^2 - 2\sqrt{s}E_Z. \quad (\text{G.19})$$

G.4 SMEFT expansion and fit structure

$$\mathcal{L}_{\text{SMEFT}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{C_i}{\Lambda^2} \mathcal{O}_i + \dots, \quad (\text{G.20})$$

$$\mathcal{A} = \mathcal{A}_{\text{SM}} + \sum_i \frac{C_i}{\Lambda^2} \mathcal{A}_i, \quad (\text{G.21})$$

$$\sigma = \sigma_{\text{SM}} + \sum_i \frac{C_i}{\Lambda^2} \sigma_i^{\text{int}} + \sum_{ij} \frac{C_i C_j}{\Lambda^4} \sigma_{ij}^{\text{quad}}, \quad (\text{G.22})$$

$$\chi^2(\vec{C}) = (\vec{O}_{\text{exp}} - \vec{O}_{\text{th}}(\vec{C}))^T V^{-1} (\vec{O}_{\text{exp}} - \vec{O}_{\text{th}}(\vec{C})), \quad (\text{G.23})$$

$$\delta_{\text{tot}}^2 = \delta_{\text{stat}}^2 + \delta_{\text{syst}}^2 + \delta_{\text{th}}^2 + \delta_{\text{param}}^2. \quad (\text{G.24})$$

G.5 Top FCNC partial widths and yields

$$N_S = \sigma_S \mathcal{L} \epsilon_S, \quad N_B = \sigma_B \mathcal{L} \epsilon_B, \quad (\text{G.25})$$

$$Z_A = \sqrt{2 \left[(S+B) \ln \left(1 + \frac{S}{B} \right) - S \right]}, \quad Z_{S/\sqrt{B}} \simeq S/\sqrt{B}, \quad (\text{G.26})$$

$$\Gamma(t \rightarrow q\gamma) = \frac{\alpha}{4} m_t (|\lambda_L|^2 + |\lambda_R|^2), \quad (\text{G.27})$$

$$\Gamma(t \rightarrow qZ)_\kappa = \frac{\alpha}{32s_W^2 c_W^2} \frac{m_t^3}{m_Z^2} (1 - r_Z)^2 (2 + r_Z) (|\kappa_L|^2 + |\kappa_R|^2), \quad (\text{G.28})$$

$$\text{Br}(t \rightarrow qX) \simeq \Gamma(t \rightarrow qX) / \Gamma_t^{\text{SM}}, \quad r_Z = m_Z^2 / m_t^2, \quad (\text{G.29})$$

$$\mathcal{L}(\mu, \theta) = \prod_{c,b} \text{Pois}(n_{cb} | \mu s_{cb}(\theta) + b_{cb}(\theta)) \prod_k \pi_k(\theta_k), \quad (\text{G.30})$$

$$q_\mu = -2 \ln \frac{\mathcal{L}(\mu, \hat{\theta}_\mu)}{\mathcal{L}(\hat{\mu}, \hat{\theta})}, \quad CL_s = \frac{p_{s+b}}{1 - p_b}. \quad (\text{G.31})$$

Part III

Glossary and references

Appendix H

Glossary

Term	Explanation
FCC	Future Circular Collider.
FCC-ee	Electron–positron stage of the FCC integrated programme.
FCC-hh	Proton–proton stage of the FCC integrated programme.
HL-LHC	High-Luminosity Large Hadron Collider.
Tera-Z	FCC-ee Z-pole programme with about 10^{12} Z bosons.
EWPO	Electroweak precision observable.
SMEFT	Standard Model Effective Field Theory.
EFT	Effective Field Theory.
BSM	Beyond the Standard Model.
FCNC	Flavour-changing neutral current.
GIM	Glashow–Iliopoulos–Maiani suppression mechanism.
Higgsstrahlung	Associated production $e^+e^- \rightarrow ZH$.
Recoil mass	Mass inferred from the system recoiling against the reconstructed Z boson.
VBF	Vector-boson fusion.
WW threshold	Energy region near $e^+e^- \rightarrow W^+W^-$ threshold.
aTGC	Anomalous triple-gauge coupling.
IDEA	Innovative Detector for Electron-positron Accelerators.
EDM4HEP	Event Data Model for High-Energy Physics.
FCCSW	FCC software framework.
FCCAnalyses	FCC analysis framework for processing EDM4HEP samples.
UFO	Universal FeynRules Output model format.
CL_s	Modified frequentist method for exclusion limits.
Asimov dataset	Representative dataset equal to the expectation, used for expected sensitivities.

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